Free AI / ML Learning Resources

A beginner-to-advanced roadmap using free YouTube channels and online platforms

1. Python first (if you need a refresher)

Resource	What it covers
freeCodeCamp: Python for Everybody (YouTube)	Full beginner Python course.
CS50P by Harvard (edX / YouTube)	Strong Python fundamentals, free to audit.

2. Math basics (only what you need)

Resource	What it covers
3Blue1Brown (YouTube)	Visual explanations of linear algebra and neural networks.
StatQuest with Josh Starmer (YouTube)	Statistics and ML concepts in plain language.

3. Machine learning courses

Resource	What it covers
Machine Learning Specialization, Andrew Ng (Coursera)	Most recommended starting point. Audit free; certificate is paid.
Google Machine Learning Crash Course	Free, hands-on and beginner friendly.
Kaggle Learn	Short free micro-courses on Python, Pandas, ML and deep learning, with practice notebooks.
fast.ai: Practical Deep Learning for Coders	Free, project-focused. Best once you know Python.

4. Deep learning and neural networks

Resource	What it covers
Deep Learning Specialization, Andrew Ng (Coursera)	Audit for free.
Andrej Karpathy: Neural Networks, Zero to Hero (YouTube)	Excellent but more advanced.

5. Generative AI and LLMs

Resource	What it covers
DeepLearning.AI short courses	Free one-hour courses on prompt engineering and building with LLMs.
Google Cloud Skills Boost: Generative AI path	Free introductory courses.
Hugging Face courses	Free hands-on NLP and LLM material.

6. Hindi / Telugu-friendly YouTube channels

Resource	What it covers
CampusX	Hindi, detailed ML playlists.
Krish Naik	English/Hinglish, project based.
Codebasics	Clear beginner-friendly ML.

Suggested learning path

1. Python basics and Pandas/NumPy (Kaggle Learn).
2. Andrew Ng's ML Specialization or Google's ML Crash Course.
3. Build 2-3 small projects (e.g. house price prediction, spam detection) and upload them to GitHub.
4. Take a generative AI short course from DeepLearning.AI.

Note: free access and course availability change over time, so check each site's current terms before starting.